

The Slop Problem

AI-Generated Pollution and the Case for Execution-Time Governance

A White Paper from Hollow House Institute

Author: Amy | Founder, Hollow House Institute

Date: May 2026

Category: Governance Research | Information Ecosystems

Institution: Hollow House Institute (HHI) | Arlington, Texas

This paper represents the independent research and policy perspective of Hollow House Institute. It is not sponsored by, affiliated with, or reviewed by any AI developer, platform company, or government agency.

"What We're Actually Talking About"

Let's start with the word itself. *Slop*. It emerged in online communities around 2023 and 2024 — not from a policy body, not from a tech company's PR department, but from the people who were already wading through it. Merriam-Webster named it their Word of the Year for 2025. That matters. It means the experience of encountering low-quality, mass-produced AI-generated content had become widespread and recognizable enough to need its own name. We had reached the point where a word for the phenomenon was necessary, and the internet collectively produced one.

Slop, in the sense I mean it here, is not simply content made with AI assistance. I want to be precise about this distinction because it is the load-bearing wall of everything that follows. Slop is AI-generated output that has been deployed into the information environment without meaningful human review, judgment, or accountability. It is the blog post written, published, and indexed by an automated pipeline before any human being has read it. It is the fake academic

abstract that circulates as a real citation. It is the five hundred SEO-optimized articles produced overnight by a content farm running a single prompt in a loop. It is the social media reply crafted by a bot and posted at scale. It is content that looks like information but is not traceable to any accountable human mind.

What makes slop specifically dangerous is not that it is always wrong. It is that it is plausible. It passes visual inspection. It has the form of knowledge — citations, structure, confident prose — without necessarily having the substance. A human being who writes something wrong can be questioned, corrected, held to account. A slop pipeline has no such accountability surface. It just keeps generating.

I want to say clearly: this is not an anti-AI paper. I use AI tools in my own research process. HHI is not arguing that generative AI is inherently harmful or that its development should be stopped. What I am arguing is that AI output is not, by default, finished work. It is a draft — a powerful, often impressive draft — that requires human judgment before it becomes part of the shared information environment. The problem we face is not that AI exists. The problem is that we have built systems that treat AI output as a finished product and deploy it at a scale and speed that makes accountability structurally impossible.

That is the slop problem. And it is accelerating.

"Why It's Accelerating: The Incentive Stack"

If you want to understand why slop proliferates, you have to look at the incentive architecture underneath it. This is not a story about bad actors alone, though bad actors absolutely exist and take full advantage of the conditions. It is a story about structural incentives that make slop the path of least resistance — economically, operationally, and technically — for a wide range of players, from individual content creators to major platforms.

Start with platform capitalism's fundamental dynamic: platforms reward volume and engagement, not accuracy or quality. An algorithmically amplified piece of content doesn't get surfaced because it is well-sourced or carefully reasoned. It gets surfaced because it generates clicks, watch time, shares, and reactions. In that environment, more content is almost always better than better content. The incentive runs directly toward production at scale, and generative AI is the most efficient production mechanism ever invented for text, image, audio, and video.

Generative AI removed the last meaningful friction point in content creation: the human. Before large language models became commercially accessible, producing content at scale required writers, editors, designers, or at minimum a coherent content strategy. The labor cost created a natural floor. Now that floor has essentially collapsed. The marginal cost of producing one more piece of content — one more article, one more summary, one more social post — has dropped to near zero. When the cost curve collapses that completely, volume explodes. And volume without accountability is slop.

Search engine optimization creates direct financial incentives for flooding the internet with AI text. A well-understood pattern in SEO — producing high volumes of keyword-targeted content to capture long-tail search traffic — has been dramatically amplified by generative AI. What previously required a content team now requires a prompt and an API key. Entire networks of AI-generated fake news sites and content farms have emerged specifically to monetize this dynamic, producing text at industrial scale with no human authorship, no editorial oversight, and no accountability when the content is false.

Social platforms are now institutionalizing slop pipelines at the product level. Features that auto-generate captions, summaries, replies, and recommended content — built into the platform itself — normalize the idea that AI-generated output is a direct substitute for human expression. When the platform itself produces AI content by default, the distinction between human and machine authorship becomes harder to perceive and easier to dismiss.

HHI Principle

The problem is not that AI removes friction. The problem is that the friction it removes was doing governance work — creating natural checkpoints where human judgment entered the process. When those checkpoints disappear, accountability disappears with them.

Bad actors understand this dynamic clearly. AI-normalized content — content that looks and sounds like everything else on the internet because it was generated by the same models — makes targeted disinformation harder to identify. When slop is everywhere, the adversarial use of AI-generated content blurs into the background noise. The normalization of AI-generated text, images, and audio creates a kind of epistemic camouflage. This is not a speculative future concern. Influence operations, coordinated inauthentic behavior networks, and state-sponsored disinformation campaigns are already exploiting the conditions that slop normalization creates.

Cory Doctorow's concept of "enshittification" is instructive here. Doctorow describes the pattern by which digital platforms gradually optimize for monetization at the expense of quality: first they are good to users, then they abuse users to serve business customers, then they abuse business customers to claw back value for shareholders. The result is a platform that has systematically hollowed itself out in pursuit of short-term extraction. The slop dynamic is enshittification at the information ecosystem level. Platforms reward volume; AI enables volume at scale; quality and accountability erode; users are left wading through an environment that looks like information but increasingly is not.

Finally, there is what I think of as the serpent-biting-its-own-tail problem. AI-generated content gets scraped back into the training data for future AI models. When slop constitutes a significant and growing share of the text on the internet, it becomes a significant and growing share of what future models learn from. Researchers have begun documenting the phenomenon of "model collapse" — the degradation of model quality that occurs when models are trained on AI-generated rather than human-generated data. The slop problem is not just an information

quality problem today. It is a structural risk to AI quality over time. We are poisoning the well that we are simultaneously drawing from.

"The Research Ecosystem Problem"

I want to dwell here for a moment, because this is where the stakes become serious in ways that extend well beyond the inconvenience of bad search results or low-quality social content.

The research ecosystem — the connected network of preprint servers, journals, citation databases, systematic reviews, literature summaries, and policy briefs that scholars, policymakers, and practitioners rely on — is not just a convenience. It is a shared infrastructure. It is the mechanism by which human knowledge accumulates, gets tested, gets corrected, and gets built upon. When that infrastructure degrades, the consequences compound across every field that depends on it. And right now, it is under serious pressure from AI slop.

Fake academic abstracts and AI-generated literature summaries are circulating as real citations. This is not a theoretical risk. Documented cases of AI-generated papers appearing in peer-reviewed journals and preprint repositories have multiplied rapidly since 2023. Some pass superficial review because they are structurally coherent — they look like real papers. The problem is that looking like a paper and being a paper are no longer the same thing. Reviewers are human, review bandwidth is finite, and the volume of AI-generated submissions is increasing faster than review capacity can scale.

Reference contamination is a related and underappreciated risk. Scholars who rely on AI-generated literature summaries to map a field may be citing sources that were never verified — or in some cases, sources that do not exist. AI systems hallucinate citations. This is well-documented. What is less often discussed is the downstream effect: when a hallucinated citation appears in a review paper, it can propagate through subsequent citations in a field, gaining apparent credibility simply through repetition. The scholar who first cited it may never have read it. The scholars who cite it subsequently may never have read the original review. The hallucination travels.

Risk Alert

If AI-generated research summaries are used as training data for future AI systems, we are not merely polluting the knowledge commons — we are building the next generation of AI on a foundation of unverifiable claims. The feedback loop is already in motion.

The epistemic load this creates for researchers is real and growing. Scholars are spending increasing proportions of their time verifying citations rather than doing original work. The verification burden is not evenly distributed — it falls disproportionately on researchers at

institutions with fewer resources, in fields where peer review infrastructure is thinner, and in interdisciplinary work where no single reviewer can command the full range of relevant expertise.

There is also what I call the homogenization paradox. AI-generated content appears diverse on the surface — different topics, different angles, different formats — but is structurally uniform at a deeper level. It is produced by systems trained on similar data, using similar generation strategies, optimized for similar measures of coherence and plausibility. The result is content that covers a lot of surface area while converging on a narrow range of conceptual framings. In research terms, this means that flooding a field with AI-generated content can actually reduce the diversity of ideas in circulation, crowding out heterodox, interdisciplinary, or qualitative perspectives that do not fit neatly into the patterns AI systems have learned to reproduce.

This is a particular threat to qualitative and interdisciplinary research, which depends on nuanced sourcing, interpretive judgment, and the kind of contextual reasoning that human scholars develop over years of immersion in a field. AI systems can produce plausible-sounding qualitative analysis. They cannot replicate the epistemic responsibility that a human researcher brings to that process — the accountability to their sources, their participants, their field, and their readers.

At HHI, we frame this clearly: the research ecosystem is shared infrastructure, and polluting it has compounding consequences that no individual researcher can remediate alone. This is a governance problem, not an individual hygiene problem.

"Provenance: The Question We Stopped Asking"

Every piece of information has an origin. That origin matters. This is not a new idea — historians, librarians, journalists, and scientists have understood provenance as a fundamental epistemic requirement for as long as those fields have existed. What is new is that we have built information systems that systematically obscure origin, and we have begun treating that obscurity as acceptable.

Provenance, in the sense I mean here, is the traceable chain from an original human thought or verified empirical event, through the tools and processes used to record and communicate it, to the published claim that appears in the information environment. A well-sourced news article has provenance: a reporter observed or verified something, wrote it under their name, with editors who reviewed it, published by an outlet with a documented editorial process. A peer-reviewed paper has provenance: researchers conducted a study, described their methods, submitted to review, revised in response to criticism, and published with their names attached. These chains are imperfect. But they are traceable. They have accountability surfaces.

AI slop severs provenance. It produces outputs that look cited and grounded — often they include in-text references, confident assertions, and structured arguments — but are not traceable to any accountable human judgment. The "hallucination" problem that AI developers discuss is, at its core, a provenance problem in disguise. When an AI system generates a plausible-sounding citation that does not exist, it is not merely making an error. It is generating the appearance of epistemic responsibility — a named source, a year, a publication — without the substance. The chain of accountability that provenance requires simply is not there.

Provenance is not a metadata field. It is the minimum condition for information to be trusted in research, policy, and public discourse.

— HHI Core Principle

Consider the contrast: a human author who publishes something wrong can be held accountable. They can be contacted, questioned, asked to respond. They can issue a correction, a retraction, or a clarification. Their institution can be notified. Their professional reputation is at stake. These are not merely social niceties — they are the accountability mechanisms that make published knowledge trustworthy enough to build on. A slop pipeline has none of these. There is no author to contact. There is no institution responsible for the claim. There is no correction mechanism. The content exists, it circulates, and the chain of accountability simply does not exist.

HHI's position is that provenance should be treated as a machine-readable, verifiable property of published information — not merely an author's note or a platform disclosure. This means generating provenance records at the moment of content creation, embedding them in a form that cannot be trivially stripped or falsified, and making them accessible to downstream consumers — whether those consumers are human readers, automated indexing systems, or AI training pipelines. Provenance is not a regulatory nicety. It is infrastructure. And right now, we are building a vast information ecosystem without it.

"Execution-Time Governance: The Missing Layer"

This is where I want to spend the most time, because this is where I think the governance conversation is most importantly incomplete — and where HHI has the most specific contribution to make.

Most AI governance discussion focuses on three intervention points: training data (what the model learns from), model safety evaluation (whether the model behaves acceptably before deployment), and deployment licensing (whether the operator has permission to use the model). These are all legitimate and important areas of governance. But they share a critical limitation:

none of them intercept the actual moment when slop is created and pushed into the information environment.

A model that passes every safety evaluation can still be used to produce slop at industrial scale. A model trained on carefully curated data can still be deployed through an automated pipeline that publishes its outputs without human review. Training-time governance and deployment licensing address whether the tool exists and whether it is safe in controlled conditions. They say almost nothing about what happens when that tool is used, at scale, in the real world, by operators whose incentives run toward volume rather than quality.

What we are missing is governance that operates at execution time — at the moment of AI output generation, as it moves from model output to published content in the information environment. I call this execution-time governance, and it is the primary governance contribution HHI is advancing in this paper.

HHI Definition

Execution-time governance refers to governance mechanisms that operate at the moment of AI output generation — intercepting the actual event at which AI-produced content enters or is prepared to enter the information environment, rather than operating only at the points of model training, safety evaluation, or deployment licensing.

What does execution-time governance look like in practice? Let me be specific, because this framework needs to be operationalizable, not just conceptually coherent.

Mandatory human-in-the-loop review checkpoints before AI-generated content is published or indexed. Not a checkbox. Not a terms-of-service agreement. An actual review event, logged and verifiable, in which a named accountable human being has evaluated the content before it enters the information environment. In high-stakes publishing contexts — academic preprints, news, government documents — this should be a hard requirement, not a recommendation.

Machine-readable provenance assertions embedded at generation time. When an AI system produces an output that will be published, the generation event should produce a provenance record — a machine-readable, tamper-evident record that includes what system produced it, when, with what inputs, and under what operator. This record should travel with the content. It should not be a post-hoc annotation that can be stripped. It should be generated at the moment of creation and preserved through the publication chain.

Platform-level throttles on AI-generated content volume per account per time period.

Volume is the mechanism by which slop does its damage. A platform that allows a single account to publish unlimited AI-generated content is a platform that has made a choice to allow unlimited slop. Rate-limiting is a blunt instrument, but it creates friction that makes automated slop pipelines less economically attractive and more visible to platform enforcement systems.

Attestation requirements for publishers. Publishers of research, news, and policy content should be required to declare whether content was AI-generated, AI-assisted, or human-authored — with legal liability attached to false attestation. This is not fundamentally different from existing disclosure requirements in other contexts (financial disclosures, ingredient labeling, country-of-origin rules). We require producers to be truthful about what they are selling in other domains. There is no principled reason to exempt information content from the same logic.

API-level logging that creates an auditable trail. AI content APIs used in high-volume publishing pipelines should be required to maintain logs of generation events — what was generated, when, by which operator, and for what stated purpose — accessible to regulators under appropriate legal process. This is not surveillance of individual users. It is the same kind of auditing we apply to other industries where the potential for systemic harm is significant.

Rate-limiting mechanisms for AI content APIs in high-volume publishing contexts. API providers who serve publishing pipelines should be required to implement and enforce rate limits that prevent automated slop factories from operating at their current scale. This does not prevent legitimate AI-assisted content creation. It creates friction specifically calibrated to the high-volume, low-accountability patterns that characterize slop production.

What This Actually Looks Like in Practice

HHI's own governance stack operationalizes each of these mechanisms. The architecture includes:

- **Append-only governance telemetry** — every governance event written to an immutable, timestamped JSONL log that cannot be retroactively altered
- **Cryptographic audit integrity** — SHA-256 checksums applied at the file level, producing tamper-evident chains across the full event history
- **Stop authority** — a human override mechanism with an explicit authority matrix that can halt AI execution at runtime, not merely flag it for later review
- **Behavioral drift monitoring** — continuous runtime observation that detects when system behavior diverges from its governance baseline, triggering alerts before drift compounds
- **Replay validation** — the ability to reconstruct and independently verify any governance event from its original inputs, confirming that logged outcomes match actual execution
- **Continuous assurance** — always-on governance processes running in parallel with operations, not applied after the fact

None of these are theoretical. HHI runs them. The point is not to showcase a stack — it is to demonstrate that execution-time governance is buildable, operable, and auditable with existing tooling. The governance gap is not a technical constraint. It is a policy choice.

The absence of these mechanisms in high-stakes publishing pipelines is not an engineering problem. Engineers could build them this week. It is a priority problem — and priority problems are governance problems. That is the intervention point.

HHI Principle

Execution-time governance is not about restricting AI capability. It is about creating accountability surfaces where currently there are none. The quality of the model is not the problem. The absence of accountability at the point of use is the problem.

I want to address the objection I expect most directly: that these measures would slow down legitimate AI-assisted content creation and impose burdens on good-faith operators. My answer is yes, they would — modestly and intentionally. The friction they introduce is not a bug. It is the point. Governance that creates no friction creates no accountability. The question is not whether to create friction, but where to create it, for whom, and with what degree of proportionality. Execution-time governance, properly designed, applies heavier friction to automated high-volume pipelines and lighter friction to individual creators who are clearly exercising human judgment. That is not an unreasonable burden. It is a reasonable design.

"Human Authorship Is Not a Luxury"

I am going to say something that will read as obvious to some people and as nostalgic or naive to others: human authorship is a functional requirement, not a cultural preference. It is not about sentiment. It is not about protecting jobs. It is not technophobia dressed up as principle. It is about what the accountability structures of a functional information environment actually require.

What does human authorship provide that AI generation does not? Let me be specific. Human authorship provides judgment under uncertainty — the capacity to make decisions about what to say and how to say it when the right answer is not computable from the available inputs. It provides accountability to a named individual or institution who can be questioned, challenged, and held responsible. It provides the ability to respond — to acknowledge error, issue correction, engage criticism, and update claims in light of new evidence. And it provides the integration of context that AI systems cannot access: lived experience, disciplinary intuition, awareness of what is at stake for actual people in an actual situation.

The distinction HHI draws is between AI as a tool used by an author, and AI as a replacement for an author. These are not the same thing, and our governance systems need to treat them differently.

Mode	Description	Accountability Surface	HHI Assessment
AI-Assisted Authorship	A human sets direction, uses AI to draft or accelerate, reviews all outputs, takes responsibility, publishes under their name	Named human author, institutional affiliation, retraction mechanism	Legitimate. Should be disclosed but is not categorically problematic.
AI-Replaced Authorship (Slop)	AI generates output; no human reviews before publication; content enters information environment directly from pipeline	None. No named human, no institutional responsibility, no correction mechanism.	This is the slop pipeline. This is what governance must address.

The "veneer of coherence" problem makes this distinction especially important. AI slop is not obviously bad. It is not the broken-grammar spam of an earlier era that readers and filters could identify on sight. It is coherent, structured, and confident. It uses the formal markers of authoritative content. This means that the harm from AI-replacing-authorship is not self-correcting through reader skepticism alone. Readers cannot reliably identify slop by looking at it. That is precisely why structural governance — provenance requirements, attestation rules, execution-time checkpoints — is necessary. The human reader cannot do the job that governance needs to do.

On the question of naming: a document that lists no accountable human author is not a document in the full epistemic sense. It is a generation event. It is an output. It may be useful, may be accurate, may be impressively well-structured — but until a human being has reviewed it and attached their name to it, it lacks the accountability surface that makes published information trustworthy. This is not a romantic or sentimental position. It is a claim about what accountability requires.

HHI's view is that human authorship should be established as a legal category with associated responsibilities and protections. This means that claiming human authorship when no human reviewed the content should carry legal liability. It also means that genuine human-authored works should carry legal protections — discovery rights, attribution rights, and the right to be distinguished from AI-generated content in archiving, indexing, and citation systems. Human

authorship is not just a norm. It is the foundation of a functioning epistemic accountability system, and it should be treated accordingly in law.

"What Governance Actually Needs to Do"

Let me consolidate the argument into a clear set of policy directions. These are written in HHI's voice — not as bureaucratic boilerplate, but as actual positions we hold and are prepared to defend. Each of these reflects a specific gap in the current governance landscape that execution-time governance is designed to address.

1. Mandatory AI content provenance standards at the API and platform level.

Machine-readable, tamper-evident provenance records should be generated at the moment of AI content creation and preserved through the publication chain. These records should include the generating system, operator, timestamp, and input context. This should not be voluntary. It should be a baseline requirement for AI content APIs operating in publishing contexts, with audit rights reserved to designated regulatory bodies.

2. Human attestation requirements for research, news, and policy content. Publishers in high-stakes categories — academic journals, preprint servers, news outlets, government information portals — should be required to declare whether content is AI-generated, AI-assisted, or human-authored. False attestation should carry legal liability under existing consumer protection and fraud frameworks, with specific provisions developed for the research and news contexts.

3. Execution-time governance frameworks in high-stakes publishing pipelines. Regulatory bodies with jurisdiction over academic publishing, news media, and government information should develop specific execution-time governance requirements — mandatory human review checkpoints, logged and verifiable — before AI-generated content is published or indexed in high-stakes contexts. These requirements should be proportionate to the stakes and volume of the publishing context.

4. Platform accountability for algorithmic amplification of AI slop. Platforms that algorithmically amplify AI-generated content — promoting it in feeds, surfacing it in search results, recommending it to new audiences — should bear partial liability for downstream epistemic harms when that content is demonstrably false or unverifiable. This mirrors the logic of product liability: when a platform's systems actively distribute harmful content, the platform is not merely a passive conduit.

5. Publicly funded, independent provenance infrastructure. Provenance infrastructure should not be owned by AI companies or platform operators. The entities with the greatest financial interest in obscuring provenance should not be the ones building the systems that

verify it. Public investment in open, auditable provenance infrastructure — similar to the logic behind public investment in internet security standards — is a prerequisite for provenance systems that can be trusted.

6. A protected legal category for human-authored works. Establish in law a distinct category for human-authored content, with associated rights: the right to accurate attribution, the right not to be misrepresented as AI-generated, and discovery rights in legal and regulatory proceedings. This category should have clear definitional criteria — including the requirement that a named accountable human reviewed the content before publication.

7. Auditable generation logs for AI content APIs. AI content APIs used in publishing contexts should be required to maintain auditable logs of generation events — operator, timestamp, volume, and stated use case — accessible to designated regulatory authorities under appropriate legal process. This is standard auditing practice in other regulated industries. There is no principled reason to exempt AI publishing infrastructure from it.

8. Investment in epistemic resilience education. Media literacy, research literacy, and information provenance education should be treated as a public infrastructure investment — not an elective curriculum add-on. This means sustained, funded integration into K-12 and higher education, with particular attention to provenance verification skills, citation hygiene, and the capacity to distinguish AI-assisted from AI-generated content. Governance structures that center human accountability need a citizenry capable of demanding and exercising that accountability.

These eight recommendations form a coherent framework rather than a menu of isolated interventions. Provenance standards without attestation requirements are unenforceable. Attestation requirements without platform accountability create perverse incentives. Platform accountability without independent provenance infrastructure hands too much power to the platforms. Epistemic resilience education without legal protections leaves citizens informed but unprotected. The recommendations work together, and governance that addresses only one or two of them while ignoring the others will find the slop problem routing around whatever barriers it encounters.

"Where HHI Stands"

I want to close in my own voice, without the policy-paper register, because I think this moment calls for directness.

HHI is not a large institution. We are not a well-funded think tank with a roster of fellows and a suite of offices. We are a focused research and policy voice, based in Arlington, Texas, working on these questions because we believe they matter — because the alternative to naming and

building frameworks for this is watching the information environment continue to degrade in ways that have real consequences for real people.

The slop problem is not going to solve itself. The incentive stack runs the other direction. Platform capitalism rewards volume. Generative AI enables unlimited volume at near-zero cost. Bad actors exploit the resulting confusion. And the research and policy ecosystems that might otherwise surface these problems are themselves increasingly contaminated by the thing they are trying to study. The conditions for self-correction are not in place. They have to be built deliberately.

What gives me some optimism — and I want to be honest that it is a measured optimism, not a triumphant one — is that the frameworks for this kind of governance already exist in other domains. We know how to require disclosure. We know how to assign liability. We know how to build auditing systems. We know how to fund public infrastructure. None of what I am proposing here requires inventing governance from scratch. It requires applying existing governance logic to a new context, with appropriate attention to what makes AI-generated content different from previous information challenges. The knowledge exists. The gap is political will and institutional coordination.

HHI's commitment is to keep naming what is happening, building frameworks that can actually be operationalized by policymakers and practitioners, and advocating for governance structures that center human accountability — not as a nostalgic preference, but as a functional requirement for a trustworthy information environment.

The slop problem deserves a serious governance response. It has not received one yet. This paper is an attempt to advance that conversation — to be specific about what execution-time governance means, why it is necessary, and what it would look like in practice.

This paper is not the end of a conversation. It is an opening move.

Amy

Founder, Hollow House Institute

Arlington, Texas

May 2026

[hollow.house](#) | Inquiries and correspondence welcome through the Institute's website.

This paper is published under Hollow House Institute's open research commitment. Citation and adaptation with attribution are encouraged. When citing, please reference: Amy. "The Slop Problem: AI-Generated Pollution and the Case for Execution-Time Governance." Hollow House Institute, May 2026.

© 2026 Hollow House Institute. All rights reserved. Published May 2026. | Arlington, Texas | hollow.house